

Techno-Economic Analysis of Floating Offshore Wind Energy vs Solar Farm to address the energy problem of Panay Island

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Abstract— The recent energy crisis in Panay Island, Philippines, resulting from a 452-megawatt loss on the Visayas grid, caused widespread outages and economic damages exceeding ₱3.8 billion in Iloilo. This event highlights the need for improved grid resilience through renewable energy. This study addresses a significant research gap by performing a comparative techno-economic analysis of floating offshore wind energy and solar PV farms. Key focus areas include Annual Energy Production (AEP), Capacity Factor (CF), Net Present Value (NPV), Internal Rate of Return (IRR), and Levelized Cost of Energy (LCoE), using RETScreen Expert for wind and the System Advisory Model (SAM) for solar PV. Data reveals that the offshore wind farm significantly outperforms the solar PV in AEP, achieving 169,151 MWh compared to 96,257 MWh and has a higher CF of 38.60% versus 21.91%. However, the solar PV boasts a higher NPV of PHP 628,979,815 compared to PHP 58,301,292 for wind, and a comparable IRR of 26.34% versus 26.70%. Despite financial advantages for solar, the wind farm's LCoE at PHP 0.102 per kWh is lower than solar's PHP 1.63 per kWh, making it more cost-effective per unit of energy. Thus, offshore wind offers a more viable solution for enhancing Panay Island's energy security and sustainability.

Keywords—Solar PV Farm, Floating Offshore Wind Farm, Techno-economic Analysis, System Advisory Model, RETScreen

I. INTRODUCTION

Energy insecurity in the Philippines stems from a confluence of factors. Limited reserve margins during peak demand periods leave the grid vulnerable to unexpected disruptions. Frequent and prolonged power outages, evidenced by high system average interruption frequency index (SAIFI) and system average interruption duration index (SAIDI), underscore the fragility of the energy supply. While some efficiency improvements have been made, the country remains on a trajectory of high and increasing per capita CO₂ emissions, highlighting its reliance on fossil fuels. Furthermore, the slow transition to renewable energy sources exacerbates this vulnerability by perpetuating dependence on volatile and potentially unreliable energy sources [1][2].

As an archipelago, the Philippines faces unique energy security challenges [3]. To address these, a strong focus is needed on developing island-based solutions that improve grid resilience and prevent widespread power outages. A recent massive power outage crippled Panay Island, plunging all major cities and provinces into darkness. This crisis stemmed from the loss of approximately 452 megawatts from the Visayas grid, attributed to factors such as maintenance shutdowns and internal problems at the Panay Energy Development Corporation (PEDC). Critically, the grid lacked sufficient reserve capacity to compensate for this sudden power loss. This energy deficit triggered a cascade of consequences, including severe economic repercussions [4]. In Iloilo alone a major part of Panay Island, estimates suggest losses exceeding ₱3.8 billion due to disrupted business operations and decreased productivity [5].

Addressing these challenges, renewable energy emerges as an appealing solution for providing efficient and sustainable electricity while mitigating environmental degradation and reducing carbon emissions. [6] Driven by the need to ensure adequate energy reserves to meet power demand, Panay Island is increasingly recognizing solar and wind energy as promising sources for its energy transition.

Among the available renewable energy technologies, floating offshore wind energy and solar photovoltaic (PV) farms offer promising renewable energy solutions, but each comes with its own advantages and challenges. Floating offshore wind energy has significantly higher energy generation potential due to stronger and more consistent wind resources, especially in deeper waters. The Philippines, with an estimated offshore wind potential of 178 GW, is actively exploring this technology [7],[8]. However, the high initial capital costs, complex installation processes, and the need for specialized infrastructure remain considerable challenges [7]. In contrast, solar photovoltaic (PV) farms are highly cost-effective, benefiting from substantial reductions in technology costs and scalability for various applications. Given the Philippines' immense solar energy potential,

solar photovoltaic (PV) farms provide a viable path toward a sustainable and secure energy future [9]. Despite these benefits, solar photovoltaic (PV) farms face issues such as intermittent energy generation due to reliance on sunlight and the substantial land requirements for large-scale deployments [9]. Despite their differences, both technologies offer valuable contributions to renewable energy generation in Panay Island, but they require careful consideration of their technical feasibility and economic viability.

Therefore, this study aims to perform a preliminary techno-economic analysis of floating offshore wind energy using RETScreen Expert and solar PV farm systems using the System Advisory Model (SAM) in Panay Island. The objective is to identify the most suitable renewable energy option for meeting reserve power requirements in the Visayas grid. The analysis evaluates key metrics, including Annual Energy Production (AEP), Capacity Factor (CF), Internal Rate of Return (IRR), Net Present Value (NPV), and Levelized Cost of Energy (LCOE).

II. METHODOLOGY

I. Pre-design

A. Area Identification

This study identifies optimal areas using Multi-Criteria Decision-Making (MCDM), a method designed to evaluate multiple, often conflicting criteria to determine the most suitable locations [10]. Specifically, MCDM was applied to develop solar and wind energy potential maps by integrating relevant criteria into the analysis. QGIS, an open-source geographic information system software [11], was used for spatial data processing and visualization. This enabled precise mapping and thorough assessment of potential sites for solar and wind energy development.

TABLE I. SOLAR FARM PARAMETER FACTORS AND CRITERIA USED

Factors	Criteria		Sources
	Unsuitable	Suitable	
Solar Irradiance (GHI)		<1300 kWh/m ²	[12]
Temperature		15° C to 40° C	
Elevation		10 m to 50 m	QGIS Built-in Analysis
Slope		5° to 15°	
Aspect		112.5 to 247.5 (South, Southeast, Southwest)	
Land Cover and Used	Water, Trees, Flooded Vegetation,	Crops, Bare Ground, Rangeland	[13]

	Built Area, Clouds		
Protected Areas	Terrestrial and Inland Waters Protected Area Coverage		[14]

TABLE II. OFFSHORE WIND FARM PARAMETER FACTORS AND CRITERIA USED

Factors	Criteria		Sources
	Unsuitable	Suitable	
Wind Speed		> 5 m/s	[15]
Wind Power Density		>500 W/m ²	
Bathymetry		-1000 m to -20 m	
Ship Density	> 2,000,000 (AIS Positions transmitted annually)		[16]
Port	Fishing Port, Shipping Port		[17]
Protected Areas	Marine Protected Area Coverage		[14]

Tables I and II present the key parameter factors and criteria utilized to identify the optimal locations for the deployment of a Solar Farm and Floating Offshore Turbines. These tables serve as a comprehensive reference for evaluating site suitability based on critical technical, and environmental considerations, ensuring a systematic approach to selecting ideal deployment areas.

II. Design

The design specifications, as required for the simulation, are detailed in the tables below to test and evaluate the techno-economic analysis of each renewable energy system.

A. Floating Offshore Wind Farm

Table III shows the chosen turbine model for the analysis obtain from RETScreen Expert's wind turbine database [19].

TABLE IV. SPECIFICATIONS OF WIND TURBINES

Device Type	Brand Name	Model	Specifications
Wind Turbine	Vestas	V90-2.0MW-80m	- Power Capacity: 50 MW - Hub Height: 80 m - Rotor Diameter/Turbine: 90 m

			- Swept Area/Turbine: 6,361.73 m ² - Shape Factor: 2
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B. Solar Farm

Table III shows the specification of the chosen model of solar PV panels and inverters [18].

TABLE III. SPECIFICATIONS OF SOLAR PANEL AND INVERTERS

Device Type	Brand Name	Model	Specification
Solar Panel Module	SunEvo	SE5-72-570MB	- Maximum power output: 570.078 W - Maximum power voltage: 41.4 V - Maximum power circuit: 13.8 A - Open-circuit voltage: 50.2 V - Short-circuit current: 14.2 A - Module Efficiency: 22.10% - Module Area: 2.580 m ²
Inverter	General Electric	RIU-2500-U	- Maximum power rating: 2750 kVA at 40°C - AC operating voltage range: 550 VAC - Wide input voltage range: 800-1500 VDC - Maximum DC current: 3508 A - Maximum efficiency rating: 98.73%

III. Development

A. Floating Offshore Wind Farm System

- Number of Wind Turbines

$$No. of Turbines = \frac{Power Capacity (MW)}{Capacity per unit (\frac{MW}{unit})} \quad (5)$$

The formulas above will result in the input parameters required for computation in each simulation tool to be used for each renewable energy system.

B. Solar PV Farm System

- Number of Solar Panels

$$No. of Panels = \frac{Total Power Capacity (MW)}{Panel Watt Rating per unit (MW/unit)} \quad (1)$$

- Solar Panels Connection

$$No. of Series_{(PV)} = \frac{V_{max}}{V_{oc}} \quad (2)$$

$$No. of Strings_{(PV)} = \frac{Total no. of PV panels}{No. of Series_{(PV)}} \quad (3)$$

- Number of Inverters

$$No. of Inverters = \frac{Total PV Power}{Inverter Maximum Power} \quad (4)$$

This section discusses the software tools used for simulating solar and wind farms to determine the output parameters essential for the techno-economic analysis.

IV. Simulation and Testing

A. System Advisor Model (SAM)

This study employs the Software Advisor Model (SAM) to conduct a thorough techno-economic analysis of solar PV systems for a solar farm on Panay Island. SAM will guide site selection, design parameters like module number and inverter sizing, and integrate system losses and weather data to estimate power output [18]. Economically, SAM calculates key performance indicators such as Installed Capacity, Annual AC Energy Output, Internal Rate of Return (IRR), Net Present Value (NPV), and Levelized Cost of Energy (LCOE), essential for evaluating the economic viability of solar farm options [18].

B. RETScreen Expert

RETScreen Expert is a software tool used for analyzing the viability and performance of wind farm projects. Technical assessment parameters include wind speeds, capacity factors, turbine specifications, and annual electricity production. Economic assessment parameters cover initial and operational costs, Net Present Value (NPV), Internal Rate of Return (IRR), payback periods, and electricity production costs [19].

C. Input Parameters

Shown in Tables V and VI are the required input parameters in each simulation tool (SAM and RetScreen) for each renewable energy system (Solar PV Farm and Floating Offshore Wind Farm).

Table V. SPECIFICATIONS OF SOLAR PANEL AND INVERTERS

Solar PV Farm (SAM Simulation)
Latitude
Longitude
Number of Panels
Modules per string in subarray
Strings in parallel in subarray
Ground Coverage Ratio
Number of Inverters
Axis

Table VI. SPECIFICATIONS OF SOLAR PANEL AND INVERTERS

Floating Offshore Wind Farm (RETScreen Simulation)
Latitude
Longitude
Wind Speed/annual @ 100m
Wind Turbine Model
Capacity per MW
Number of Units
Array losses
Airfoil losses
Misc losses
Availability

V. Techno-economic Analysis

The scale of 50MW power capacity of both renewable technologies will be considered as the basis for fair comparability of techno-economic analysis [20]. This analysis will provide valuable insights into the technical and economic viability of both options, enabling informed decision-making regarding the most suitable solution for meeting reserve power requirements and achieving a sustainable energy future [21], [22]. The following are common metrics to be compared for each renewable energy system.

- Annual Energy Production - measures the total energy generated by a renewable energy system over a year, typically in megawatt-hours or gigawatt-hours. A favorable AEP is not only high but also meets the energy demands it targets, considers system scale, and minimizes environmental impact. It should align with or exceed predictions based on the system's design and local conditions, be economically viable, and perform efficiently compared to similar systems [23].

- Capacity Factor - the ratio of actual energy produced to the potential output at full capacity, shown as a percentage. A high CF indicates that the system consistently operates near its maximum capacity, reflecting efficient technology use. Factors enhancing CF include appropriate technology selection for the location, effective maintenance, and reliable operation meeting demand [23].
- Net Present Value (NPV) - A positive NPV is required for a project to be profitable. The higher the NPV of the project is, the more benefit it will obtain [18], [24].

$$Net\ Present\ Value\ (NPV) = \sum_{n=0}^N \frac{F_n}{(1+d)^n} \quad (6)$$

Internal Rate of Return (IRR) - defined as a discount rate that yields an NPV of zero. A project's viability is determined by the extent to which the IRR surpasses the cost of capital [18], [24].

$$Net\ Present\ Value\ (NPV) = \sum_{n=0}^N \frac{F_n}{(1+IRR)^n} \quad (7)$$

- Levelized Cost of Energy (LCoE) - a low LCoE means the project can produce electricity more cheaply, which is attractive. If the LCoE is competitive with or lower than market electricity prices or other generation options, it suggests the project could be feasible [18], [24].

$$\sum_{n=1}^N \frac{Q_n \times LCoE}{(1+d)^n} = \sum_{n=0}^N \frac{C_n}{(1+d)^n} \quad (8)$$

III. RESULT AND DISCUSSION

I. Selected Sites

A. Solar Farm

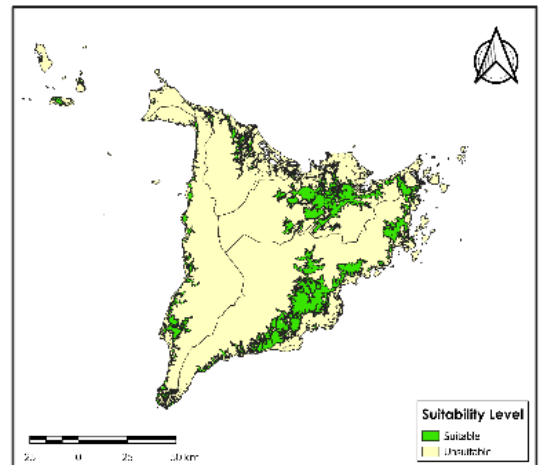


Figure 1. Suitability Level Area of Solar PV Farm

The researchers conducted a comprehensive analysis by overlaying the layers derived from all parameter criteria to evaluate the potential for Solar PV farm development. The resulting overlaid maps classified the study area into two distinct levels of suitability: SUITABLE and UNSUITABLE.

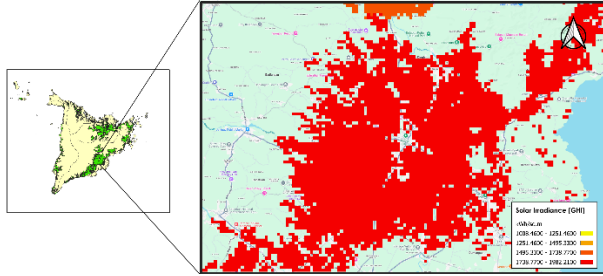


Figure 2. Study Area for Solar PV Farm

The study area encompasses an approximate area of 370,320,595.843 square meters, making it highly suitable for the establishment of a solar PV farm. Based on the specifications of the solar panel model used, which requires approximately 250,000 square meters to generate 50 MW of solar power, this area offers substantial capacity for large-scale solar energy production. Furthermore, the site exhibits excellent solar irradiance levels, averaging 1,800 kWh/m², further enhancing its suitability for solar energy generation

B. Floating Offshore Wind Farm

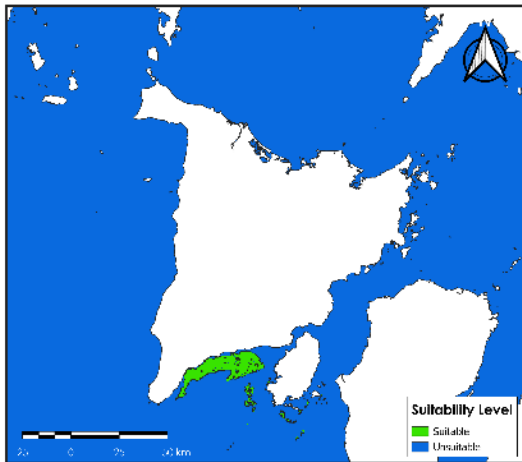


Figure 3. Suitability Level Area of Offshore Wind Farm

This map represents the suitability levels derived from the overlaid analyses from the parameter factor for offshore wind farm, specifically designed to identify optimal areas for the deployment of offshore wind farms.

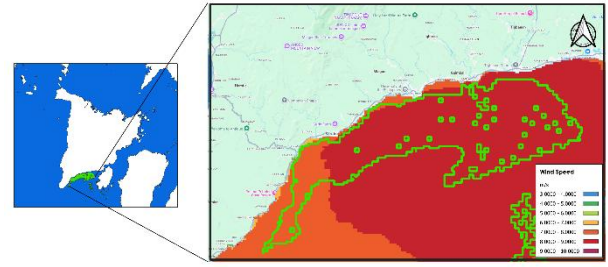


Figure 4. Study Area for Offshore Wind Farm

This is the study area with square meter. In this site we deploy 25 turbines in order to make 50 W offshore wind farm. The site also has wind speed ranging to m/s.

II. Modelling and Simulation

Table VII. INPUT VALUES FOR SOLAR PV FARM SIMULATION

Solar PV Farm (SAM Simulation)	Values
Latitude	10.937867824114509
Longitude	122.638978511029
Number of Panels	87,720
Modules per string in subarray	30
Strings in parallel in subarray	2,924
Ground Coverage Ratio	0.3
Number of Inverters	20
Axis	2

Table VIII. INPUT VALUES FOR OFFSHORE WIND FARM SIMULATION

Floating Offshore Wind Farm (RETScreen Simulation)	Values
Latitude	10.626100006939
Longitude	122.213348388672
Wind Speed/annual @ 100m	8.35 m/s
Wind Turbine Model	VESTAS V90-2.0 MW - 80m
Capacity per MW	2MW
Number of Units	25
Array losses	4%
Airfoil losses	2%
Misc losses	5%
Availability	98%

*The annual wind speed at 100 meters was obtained from the Global Wind Atlas option in RETScreen.

Table IX. SIMULATION RESULTS AND METRICS OF SOLAR PV FARM

METRIC	VALUES
Annual Energy Production	96,256.968 MWh
Capacity Factor	21.91%
Net Present Value	Php 628,979,815.05
Internal Rate of Return	26.34 %
Levelized Cost of Energy	Php 1.63 /KWh

Table X. SIMULATION RESULTS AND METRICS OF OFFSHORE WIND FARM

METRIC	VALUES
Annual Energy Production	169,151 MWh
Capacity Factor	38.6 %
Net Present Value	Php 58,301,291.9776237
Internal Rate of Return	26.7 %
Levelized Cost of Energy	Php 0.102 /KWh

III. Techno-economic Analysis

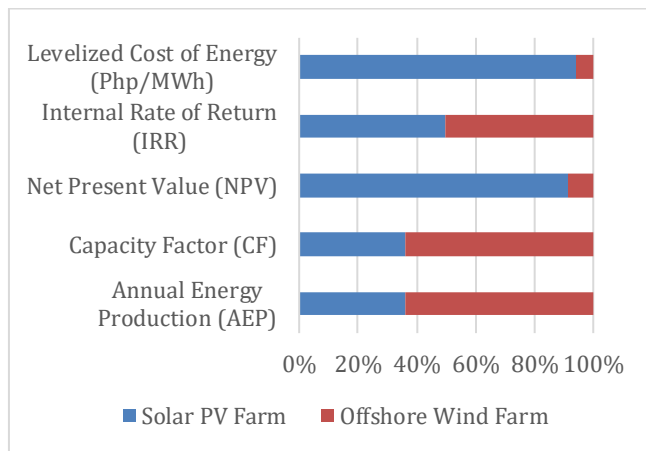


Figure 5. Comparison of Metric by Percentage

The data show that the offshore wind farm significantly outperforms the solar PV farm in Panay Island in terms of Annual Energy Production (AEP) with 169,151 MWh compared to 96,257 MWh, and Capacity Factor (CF) at 38.60% versus 21.91%. However, the solar PV farm boasts a substantially higher Net Present Value (NPV) at PHP 628,979,815 compared to PHP 58,301,292 for the wind farm, and a greater Internal Rate of Return (IRR) at 26.34% against 26.70%. Despite these financial advantages for the solar farm, the Levelized Cost of Energy (LCoE) is higher at PHP 1.63 per kWh, whereas the wind farm has a lower LCoE of PHP 0.102 per kWh, making it more cost-effective per unit of energy generated.

CONCLUSION

Given the variable energy demand and recurrent disruptions in coal-fired power plants on Panay Island, the offshore wind farm presents a more effective alternative compared to solar PV farms in bolstering grid resilience. Due to its higher capacity factor and enhanced energy output, the offshore wind farm is better suited to meet the island's peak energy requirements reliably and plays a crucial role in diversifying the energy portfolio of Panay Island, thereby enhancing its energy security and sustainability.

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REFERENCES

- [1] "Evaluating Energy Security in the Philippines: An Indicators-Based Assessment," *Pids.gov.ph*, 2019. <https://pids.gov.ph/publication/policy-notes/evaluating-energy-security-in-the-philippines-an-indicators-based-assessment>
- [2] A. Navarro, M. K. Ortiz, and J. E. Camara, "How Energy Secure is the Philippines?," Dec. 2023, doi: <https://doi.org/10.62986/dp2023.15>.
- [3] A. Syauqi, Yoga Wienda Pratama, and Widodo Wahyu Purwanto, "Sustainable Energy System in the Archipelagic Country: Challenges and Opportunities," *Green energy and technology*, pp. 49–69, Jan. 2021, doi: https://doi.org/10.1007/978-3-030-67529-5_3.
- [4] CEDTyClea, "ERC: Committee looking into Panay Island power outage - BusinessWorld Online," *BusinessWorld Online*, Jan. 04, 2024. <https://www.bworldonline.com/economy/2024/01/04/567014/erc-committee-looking-into-panay-island-power-outage/>
- [5] J. V. Ordonez, "Iloilo lost P3.8B due to blackouts - governor," *Business World*, Jan. 10, 2024. <https://www.bworldonline.com/the-nation/2024/01/10/568170/iloilo-lost-p3-8b-due-to-blackouts-governor/>
- [6] K. K. Jaiswal *et al.*, "Renewable and sustainable clean energy development and impact on social, economic, and environmental health," *Energy Nexus*, vol. 7, no. 7, p. 100118, Sep. 2022, doi: <https://doi.org/10.1016/j.nexus.2022.100118>.
- [7] "Offshore Wind Roadmap for the Philippines | Department of Energy Philippines," *Doe.gov.ph*, 2023. <https://doe.gov.ph/announcements/offshore-wind-roadmap-philippines>
- [8] National Grid, "Onshore vs offshore wind energy: what's the difference? | National Grid Group," *www.nationalgrid.com*, Mar. 30, 2022. <https://www.nationalgrid.com/stories/energy-explained/onshore-vs-offshore-wind-energy>
- [9] "IT'S MORE SUN IN THE PHILIPPINES Facts and Figures on Solar Energy in the Philippines Project Development Programme (PDP) Southeast-Asia." Available: <https://doe.gov.ph/sites/default/files/pdf/netmeter/policy-brief-its-more-sun-in-the-philippines-V3.pdf>
- [10] J. Y. L. Yap, C. C. Ho, and C.-Y. Ting, "A systematic review of the applications of multi-criteria decision-making methods in site selection problems," *Built Environment Project and Asset Management*, vol. 9, no. 4, pp. 548–563, Sep. 2019, doi: <https://doi.org/10.1108/bepam-05-2018-0078>.
- [11] "QGIS: What Is It and How Does It Work?," *xyHt*, Sep. 01, 2018. <https://www.xyht.com/spatial-itgis/qgis-what-is-it-and-how-does-it-work/>
- [12] Solargis, "Global Solar Atlas," *Globalsolaratlas.info*, 2020. <https://globalsolaratlas.info/map>

- [13] "ArcGIS Online," *Arcgis.com*, 2018. <https://www.arcgis.com/home/index.html>
- [14] Protected Planet, "Home," *Protected Planet*, 2024. <https://www.protectedplanet.net/en>
- [15] Global Wind Atlas, "Global Wind Atlas," *globalwindatlas.info*, 2024. <https://globalwindatlas.info/en/>
- [16] "Data Catalog," *Worldbank.org*, 2022. <https://datacatalog.worldbank.org/search/dataset> (accessed Jan. 22, 2025).
- [17] "GP," *www.geoportal.gov.ph*. <https://www.geoportal.gov.ph/>
- [18] A. Patricia *et al.*, "Techno-economic Feasibility Analysis of a Floating Solar Photovoltaic System in San Roque Dam, Philippines," Aug. 2023, doi: <https://doi.org/10.1109/iconnect56593.2023.10326973>.
- [19] S. S. Asamoah, J. Parbey, I. K. Yankey, and A. Awuah, "Techno-economic assessment of a central grid-connected wind farm in Ghana using RETScreen® Expert," *Heliyon*, vol. 9, no. 1, p. e12902, Jan. 2023, doi: <https://doi.org/10.1016/j.heliyon.2023.e12902>.
- [20] "Techno-Economic Analysis for Wind Energy Projects: A Comparative Study With Three Wind Turbines Based on Real-Site Data," *Turkish Journal of Electrical Power and Energy Systems*, vol. 3, no. 3, pp. 115–124, Oct. 2023, doi: <https://doi.org/10.5152/tepes.2023.23019>.
- [21] DOE Advanced Manufacturing Office, "Introduction to Techno-Economic Analysis." https://www.energy.gov/sites/default/files/2022-01/2022-01-19%20-%20Intro%20to%20TEA%20-%20Slides%20and%20Transcript_compliant_1_0.pdf
- [22] "International Best Practices in Solar and Wind Power Forecasting GET.transform Technical Brief." Accessed: Jan. 22, 2025. [Online]. Available: https://www.get-transform.eu/wp-content/uploads/2024/01/GET.transform-Brief_VRE-Forecasting-Solar-Wind.pdf
- [23] J. Timmerman, L. Vandevelde, and G. Van Eetvelde, "Towards low carbon business park energy systems: Classification of techno-economic energy models," *Energy*, vol. 75, pp. 68–80, Oct. 2014, doi: <https://doi.org/10.1016/j.energy.2014.05.092>.
- [24] J. Zeng *et al.*, "Environmental, Energy, and Techno-Economic Assessment of Waste-to-Energy Incineration," *Sustainability*, vol. 16, no. 10, p. 4140, Jan. 2024, doi: <https://doi.org/10.3390/su1610414>